المكتبة الذكية
Smart Library LMS · Edition 2026

INSTITUTIONAL RELEASE V2.4

COMPREHENSIVE SYSTEM DOCUMENTATION & VISUAL WALKTHROUGH

Smart Library Management System (LMS 2026)

A complete, production-grade institutional platform for physical item circulation, AI-powered semantic discovery, priority hold queues, financial penalty settlements, and multi-tenant library governance.

FRONTEND CORE	BACKEND ENGINE	AI ASSISTANT	LENDING CURRENCY
React 19 + Vite 8	Express + MySQL2	Google Gemini 1.5	EGP (ج.م) Official

Prepared for Executive Review & Technical Operations
Full feature audit, role-based workflows, architecture schemas, and high-fidelity screen captures.

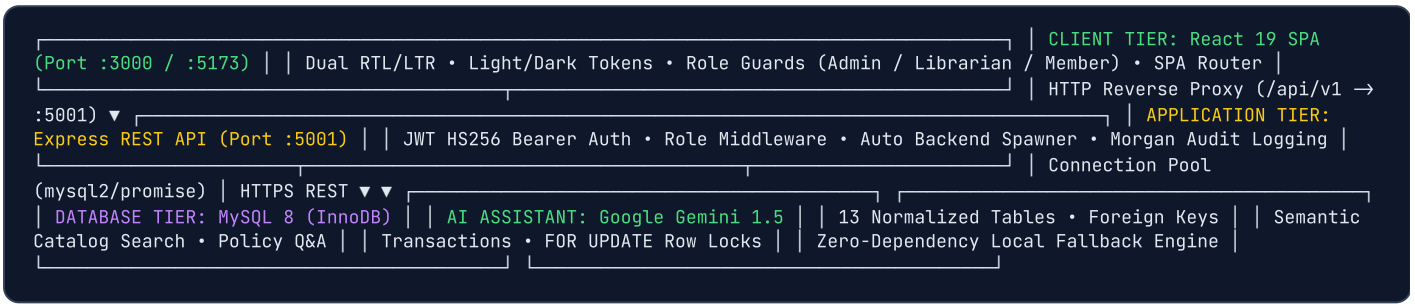
March 2026
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1. Executive Summary & System Architecture

Smart Library LMS is an academic-grade digital catalog and circulation platform engineered for institutional libraries. It solves the operational challenges of modern physical book collections: tracking distinct barcode-tagged copies, automating ACID loan checkouts and returns, computing daily late penalties in Egyptian Pounds (EGP), managing priority reservation queues with 48-hour pickup expiration, and assisting readers with natural-language discovery powered by **Google Gemini**.

CATALOG TITLES	PHYSICAL COPIES	CIRCULATION RULES	PENALTY RATE
8 Master	**27 Tracked**	**ACID Atomic**	**5.00 EGP/Day**
Bilingual metadata	Unique barcodes BC-xxx	Row locking FOR UPDATE	Automated checkin assess

System Architecture Diagram



Core Architectural Principles

1. Strict Separation of Title & Physical Copy

A master bibliographic record represents the intellectual work, while physical copies (`BC-202-01` , `BC-202-02`) represent inventory items with shelf coordinates (`Tech Wing - Shelf CS-18`) and live states (`available` , `borrowed` , `reserved` , `maintenance`).

2. ACID Concurrency & Overdue Calculation

Physical checkouts lock inventory rows with `SELECT ... FOR UPDATE` to eliminate race conditions. Checkins calculate overdue delta: `Math.ceil((NOW() - due) / DAY) * finePerDay` and append an immutable fine claim in Egyptian Pounds.

> **Dual Internationalization & Design Tokens**
> Native support for Arabic (RTL) with *Cairo* typography and English (LTR) with *Plus Jakarta Sans*. Instant dynamic direction flipping without full-page refreshes.

2. Complete Vendor & Third-Party Dependency Ecosystem

Smart Library LMS is built on an audited, production-grade dependency suite:

VENDOR / PACKAGE	VERSION	CLASSIFICATION	PURPOSE & IMPLEMENTATION SCOPE
React / React DOM	`^19.3.1`	UI Framework	Declarative components, hooks state, dynamic popstate routing, and context propagation.
Lucide React	`^1.61.0`	Iconography Vendor	40+ pixel-perfect SVG operational icons (circulation, barcodes, status shields, stars, AI).
Vite	`^8.3.4`	Bundler & Dev Server	High-speed ESM development server, HMR, and custom `autoBackendPlugin` child-process runner.
Express	`^4.21.2`	Backend Framework	HTTP router, modular domain controllers, error middleware, and CORS security pipeline.
MySQL2	`^3.12.0`	Database Driver	InnoDB connection pooling, parameterized queries, and ACID transactional guarantees.
JSONWebToken	`^9.0.2`	Authentication	Stateless cryptographic Bearer tokens (HS256) with 7-day expiration and claim validation.
BcryptJS	`^2.4.3`	Cryptography	Salted one-way hashing (work factor 10) for stored staff and member credentials.
Google Gemini API	`gemini-1.5-flash`	Cloud AI Provider	Generative conversational search, natural language book matching, and policy assistance.
Google Fonts CDN	Hosted	Typography Vendor	Academic typography: **Cairo** (weights 600–800) and **Plus Jakarta Sans** (400–800).

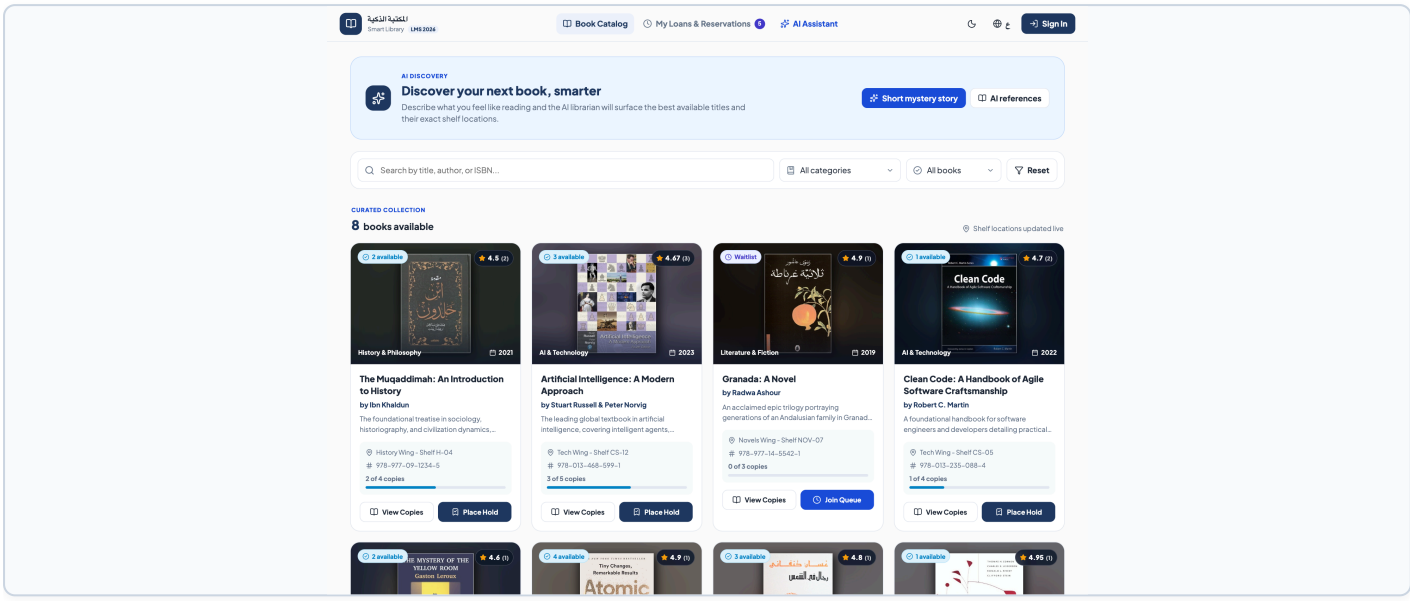
3. Role-Based Access Control (RBAC) Matrix

FUNCTIONAL CAPABILITY	PUBLIC / VISITOR	PATRON MEMBER (عضو)	LIBRARIAN (أمين المكتبة)	ADMINISTRATOR (المدير)
Browse Catalog & Shelf Mapping	ALLOWED	ALLOWED	ALLOWED	ALLOWED
Submit Star Ratings & Reviews	SIGN-IN REQ.	ALLOWED	ALLOWED	ALLOWED
AI Discovery Assistant Drawer	ALLOWED	ALLOWED	ALLOWED	ALLOWED
Place Hold / Join Waitlist Queue	SIGN-IN REQ.	ALLOWED	ALLOWED	ALLOWED
Personal Dashboard & Loan Renewal	DENIED	ALLOWED (/MY-LOANS)	N/A	N/A
Circulation Desk (Barcode Issue/Return)	DENIED	DENIED	ALLOWED	ALLOWED
Title Creation & Physical Inventory	DENIED	DENIED	ALLOWED	ALLOWED
Collect & Settle Financial Fines (EGP)	DENIED	DENIED	ALLOWED	ALLOWED
Staff Provisioning & Credential Mgmt	DENIED	DENIED	DENIED	ALLOWED (/ADMIN)
System Operational Policies Configuration	DENIED	DENIED	DENIED	ALLOWED
Immutable Compliance Audit Log	DENIED	DENIED	DENIED	ALLOWED

> **Seeded Demo Credentials:**
> **Admin:** `admin@library.sa` / `admin123` | **Librarian:** `sulaiman@library.sa` / `librarian123` | **Member:** `ahmed@library.sa` / `member123`

Walkthrough Stage 1: Public Patron Discovery & Catalog

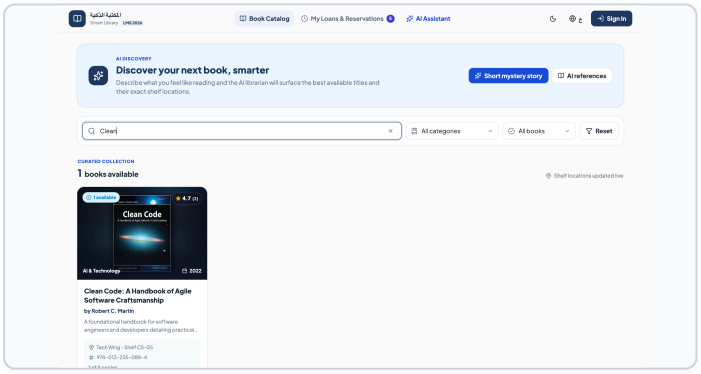
The public-facing catalog serves as the central showcase for the library. Built with responsive grid cards, it communicates bibliographic metadata, real-time physical availability, exact shelving coordinates, and reader reviews.



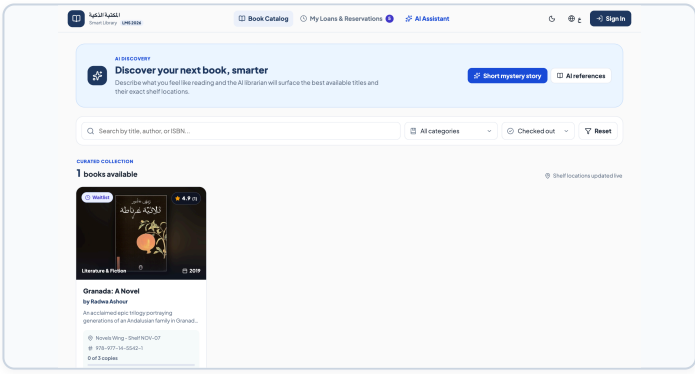
UI WALKTHROUGH Public catalog homepage displaying AI discovery banner, dynamic filter bar, curated book cards, and real-time copy availability meters.

Key Features of the Discovery Interface

- ✓ **Live Omnisearch:** Search instantly across title (Arabic & English), author, ISBN-13, and category taxonomy with zero latency.
- ✓ **Taxonomy Category Select:** Filter by *AI & Technology*, *Literature & Fiction*, *History & Philosophy*, *Mystery & Crime*, and *Self-Development*.
- ✓ **Availability Status Filter:** Instant segmentation between books currently in stock and titles with active hold waitlists.
- ✓ **Physical Location Badge:** Every card provides the exact wing and shelf code (e.g. `Tech Wing - Shelf CS-12`).



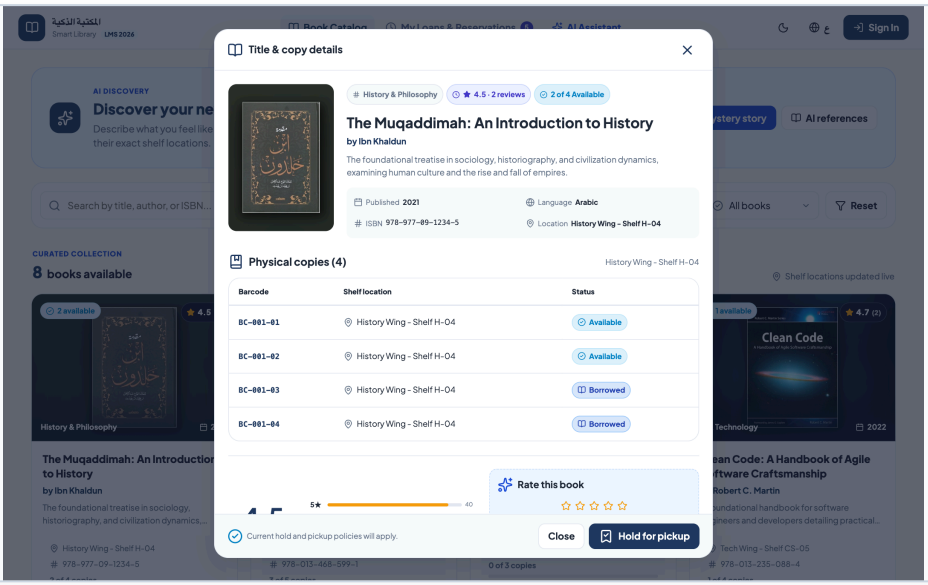
UI WALKTHROUGH Live search query filtering by keywords ('Clean') with instant card reduction.



UI WALKTHROUGH Availability filter highlighting checked-out titles with 'Waitlist' badges.

Walkthrough Stage 2: Bibliographic Record & Physical Barcodes

Clicking any book card opens the **Deep Bibliographic Modal**. Unlike traditional catalogs that hide physical inventory details, Smart Library exposes individual copy barcode tracking and community engagement.



UI WALKTHROUGH Detailed bibliographic modal showing book cover, publication year, ISBN, physical copy barcode inventory table, and hold reservation action.

Physical Copy Tracking Table

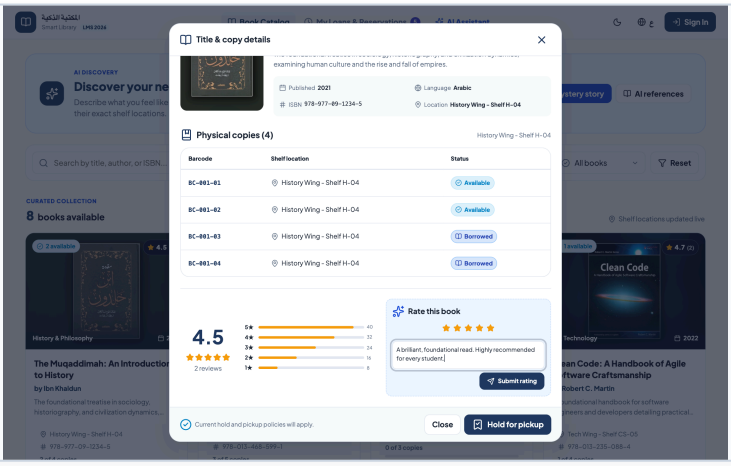
The modal displays every physical copy assigned to the title with its unique barcode (e.g. `BC-001-01`, `BC-001-02`), specific shelving location, and current state. If authenticated as staff, librarians can toggle individual copy status directly from this view.

Hold Reservation & Queueing

If copies are available, patrons can immediately place a hold. If all copies are on loan, the primary button switches to **Join Waitlist**, calculating their upcoming position in the priority queue.

Interactive Ratings & Reviews

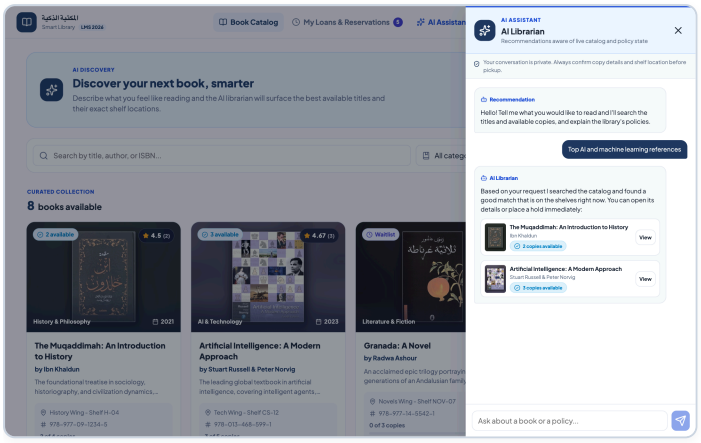
Readers view the overall rating score (e.g. 4.67★), review count, and a 5–to–1 star distribution histogram. Authenticated members can select 1–5 stars with hover animation and author a written critique.



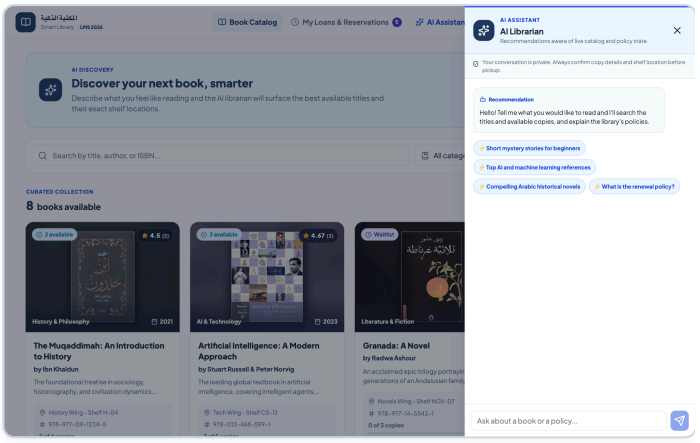
UI WALKTHROUGH Rating histogram and review submission box with interactive star rating selector and text critique.

Walkthrough Stage 3: AI Discovery Assistant (Google Gemini)

Accessible via the top navigation or banner shortcuts, the **AI Discovery Assistant** provides a slide-over conversational drawer. It leverages the `gemini-1.5-flash` model to evaluate user reading intents and suggest matching titles currently shelved in the collection.



UI WALKTHROUGH AI slide-over drawer with conversational response, semantic catalog recommendations, and direct 'View' links.



UI WALKTHROUGH Curated quick-prompt starter pills for mystery fiction, AI references, history, and lending policies.

Curated Starter Prompts

Users can click pre-seeded shortcuts to explore the catalog:

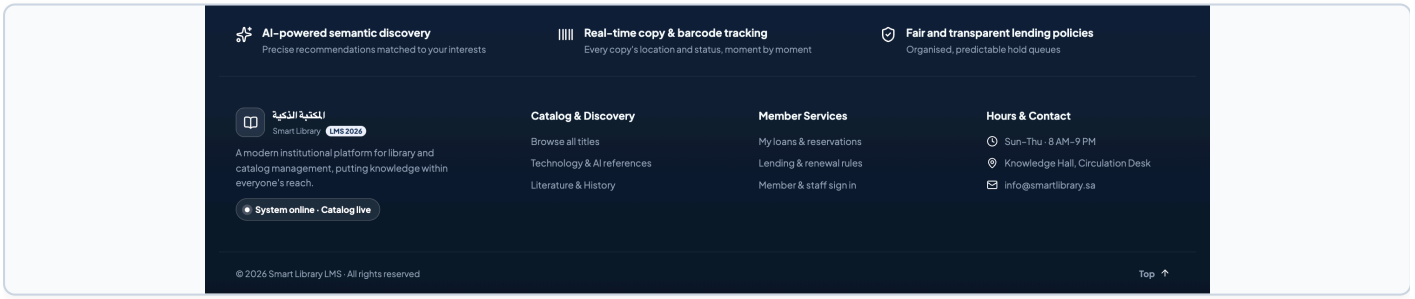
- ✓ *"Short mystery story for beginners"*
- ✓ *"Top AI and machine learning references"*
- ✓ *"Compelling Arabic historical novels"*
- ✓ *"What is the renewal and loan policy?"*

Dual Fallback Architecture

If the Google Gemini API key is absent or unreachable, the system automatically falls back to an internal keyword matching engine. Users never encounter an empty screen or broken interface.

Navy Institutional Footer (Refined Centered Alignment)

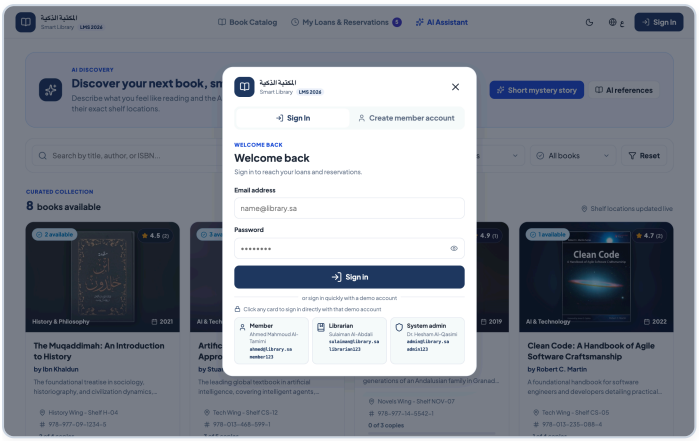
Features an executive deep navy gradient footer (`#112238` to `#0c1827`) incorporating high-contrast white icons, operating hours, real-time copy tracking guarantees, and contact metadata.



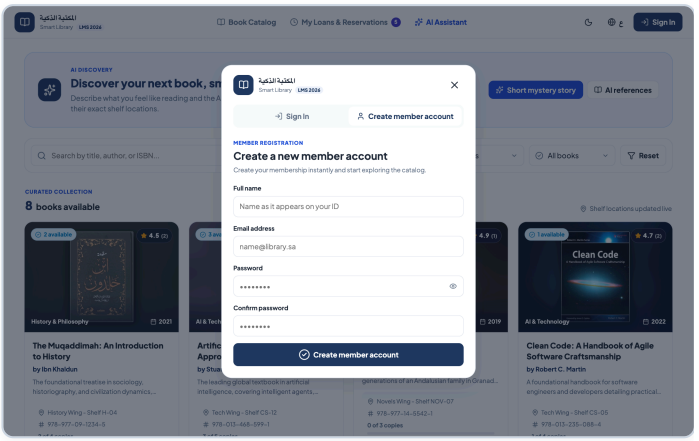
UI WALKTHROUGH Executive navy footer showcasing white centered branding icon, operational highlights, discovery links, and catalog status badge.

Walkthrough Stage 4: Authentication & Security Subsystem

Security in Smart Library LMS is built upon stateless **JSON Web Tokens (JWT)**, bcrypt credential hashing, and role-based route guards. The login overlay allows seamless authentication for patrons and institutional staff.



UI WALKTHROUGH Login modal with email/password inputs, show/hide password toggle, and one-click demo credentials cards.



UI WALKTHROUGH Member registration tab for onboarding new library patrons with password confirmation and validation.

Fast Demo Account Switcher

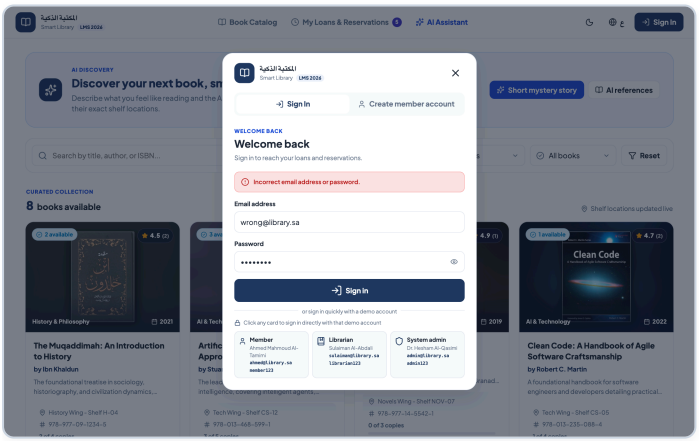
For demonstration, the modal includes interactive preset cards for **Member**, **Librarian**, and **Admin**. Clicking any preset instantly populates verified credentials and triggers authenticated sign-in.

Strict Route Protection

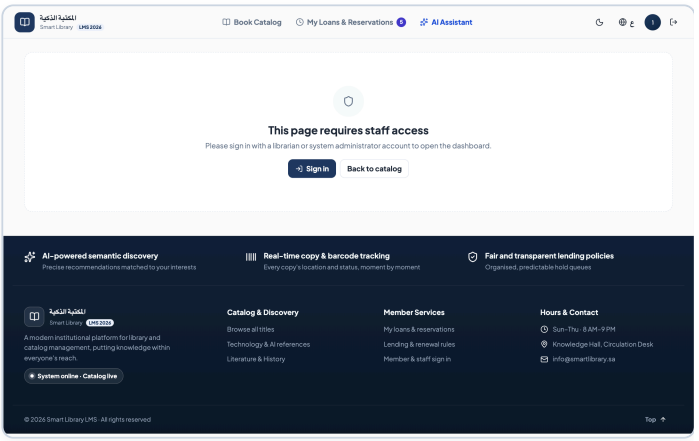
Staff dashboards (`/librarian/*` and `/admin/*`) and member portals (`/my-loans`) enforce role verification. Unauthenticated access attempts are redirected to an **Access Denied** security gate with single-click login redirect.

Robust Error Handling

Invalid passwords, deactivated accounts, or duplicate emails trigger clear alert banners with contextual advice.



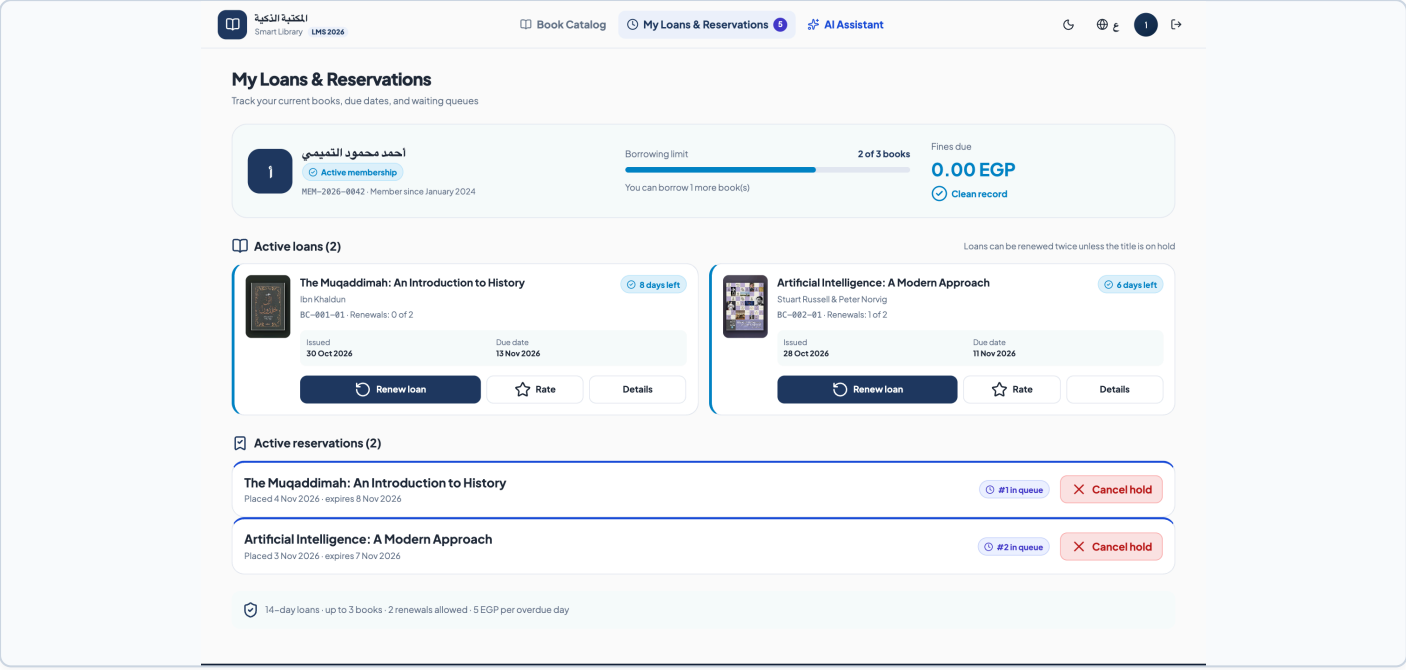
UI WALKTHROUGH Validation error banner displaying contextual feedback when credentials fail verification.



UI WALKTHROUGH Security guard screen preventing unauthorized access to staff consoles with redirect options.

Walkthrough Stage 5: Member Self-Service Portal

The `/my-loans` portal empowers registered members to monitor their borrowed books, check countdowns until due dates, execute self-service renewals, track hold positions, and confirm good financial standing.



UI WALKTHROUGH Comprehensive member dashboard with borrowing quota progress meter, active loan cards, renewal controls, and queue position tags.

Key Capabilities of the Member Portal

1. Borrowing Limit Gauge & Good Standing

A live progress meter visually communicates quota usage (e.g. *2 of 3 allowed books borrowed*). The financial status panel confirms `0.00 EGP` balance or indicates overdue settlement requirements.

2. Countdown Due Date Badges

Each active loan card displays real-time countdown tags (e.g. `8 DAYS LEFT` or `OVERDUE BY 5 DAYS`), allowing members to proactively manage returns.

3. One-Click Loan Renewal

Members can extend active loans directly by clicking **Renew Loan**, advancing the due date by 14 days without staff intervention, provided renewal limits (max 2) and hold locks are satisfied.

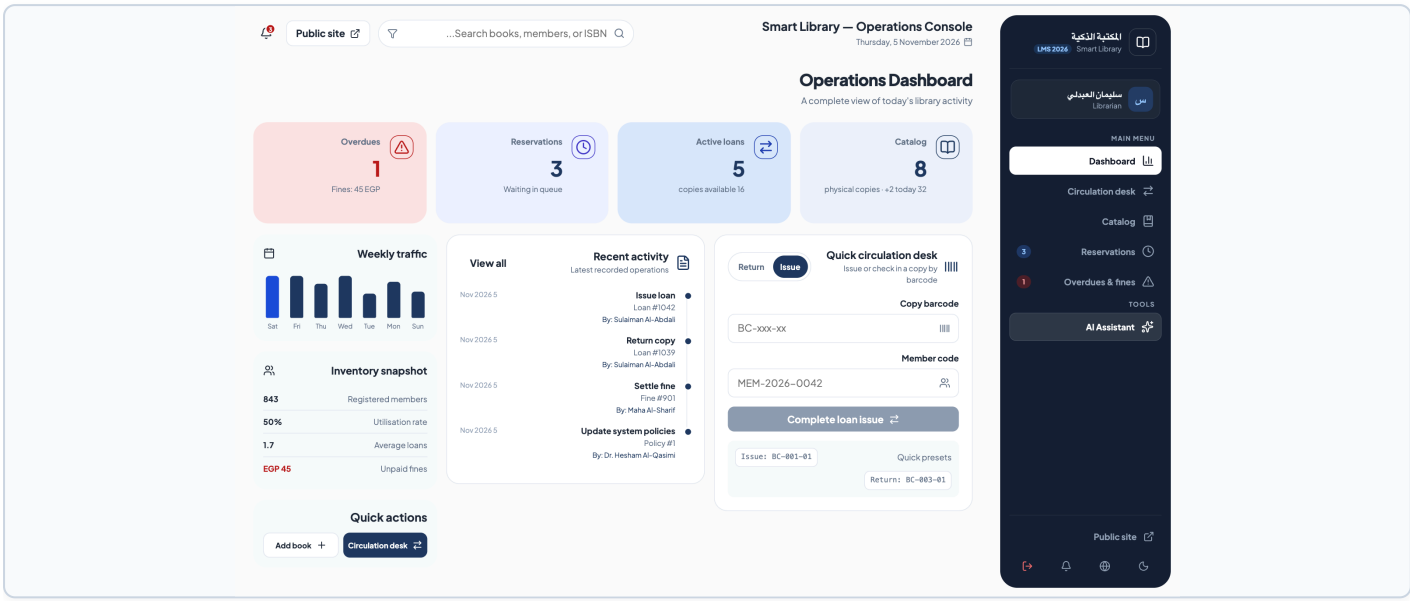
4. Hold Queue Priority Management

Displays active waitlist reservations with exact priority numbers (e.g. `QUEUE #1`, `QUEUE #2`) and self-service cancellation actions.

> **Self-Service Transparency**
> By providing members with full visibility into their borrowing standing and countdowns, the library minimizes overdue incidents and promotes responsible circulation.

Walkthrough Stage 6: Librarian Operational Dashboard

The `/librarian` console provides circulation librarians with a unified command center. It combines real-time operational KPI counters, an instant barcode circulation desk, recent activity timelines, and inventory health metrics.



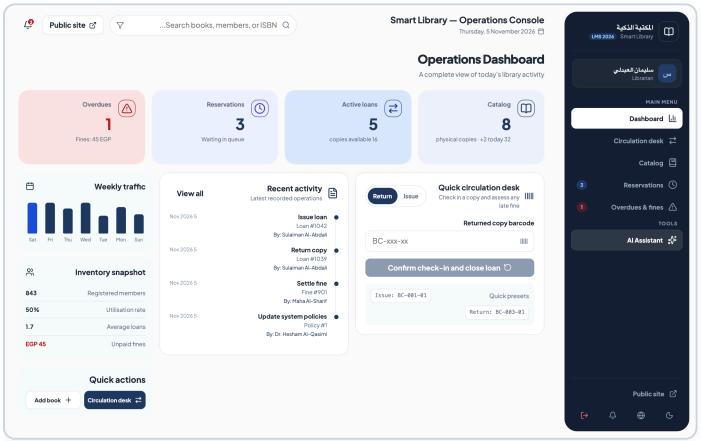
UI WALKTHROUGH Librarian operational dashboard with 4 KPI cards, barcode quick circulation desk, recent audit timeline, and weekly traffic analytics.

Operational KPI Metrics

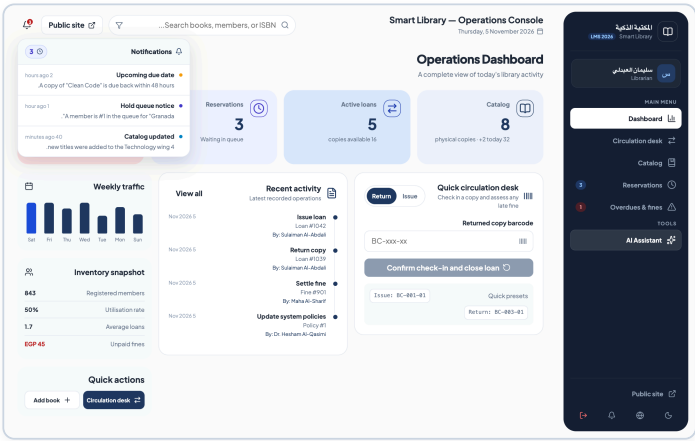
- ✓ **Catalog Volume:** Total master titles and physical copies in inventory.
- ✓ **Active Circulation:** Concurrent loans and available copies.
- ✓ **Hold Queues:** Members awaiting reserved titles.
- ✓ **Overdue Claims:** Delinquent loans and total uncollected fines in EGP.

Weekly Circulation Traffic

A normalized 7-day bar chart illustrates physical visits and checkout velocity from Sunday through Saturday, assisting staff scheduling.



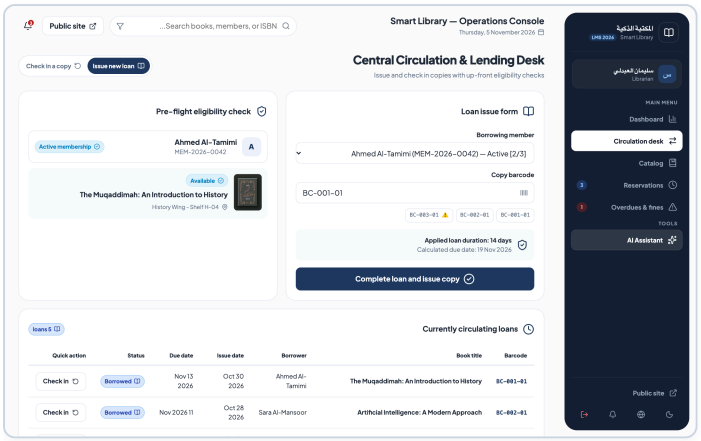
UI WALKTHROUGH Quick desk switched to Check-in mode for instantaneous barcode scanning and item return.



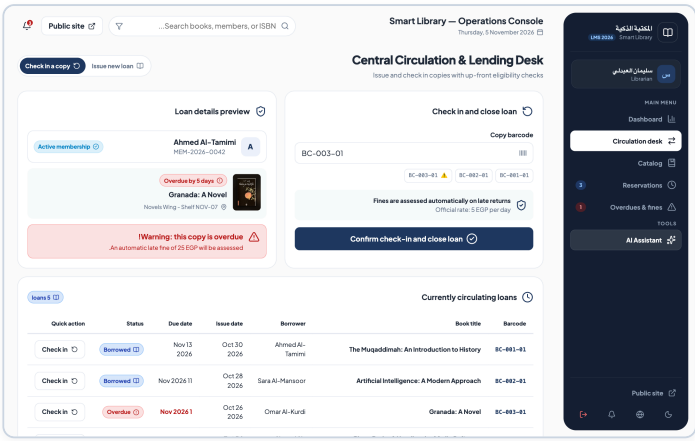
UI WALKTHROUGH Staff notification dropdown displaying real-time alerts for impending overdues and new holds.

Walkthrough Stage 7: Central Circulation & ACID Lending Engine

The dedicated Circulation Desk (`/librarian/circulation`) is the primary workhorse for physical checkouts and checkins. It performs real-time pre-flight verification before completing transactions.



UI WALKTHROUGH Circulation issue workflow: member quota check, physical copy verification, calculated due date, and atomic issue button.



UI WALKTHROUGH Return verification panel detecting an overdue return with an automatic 25 EGP fine penalty warning.

Pre-Flight Validation Pipeline

Step 1: Member Standing Inspection

Upon selecting a member (e.g. `MEM-2026-0042`), the system validates that their membership is active, current loans (`2/3`) do not exceed the policy limit, and no unpaid overdue fines exist.

Step 2: Copy Availability & Row Locking

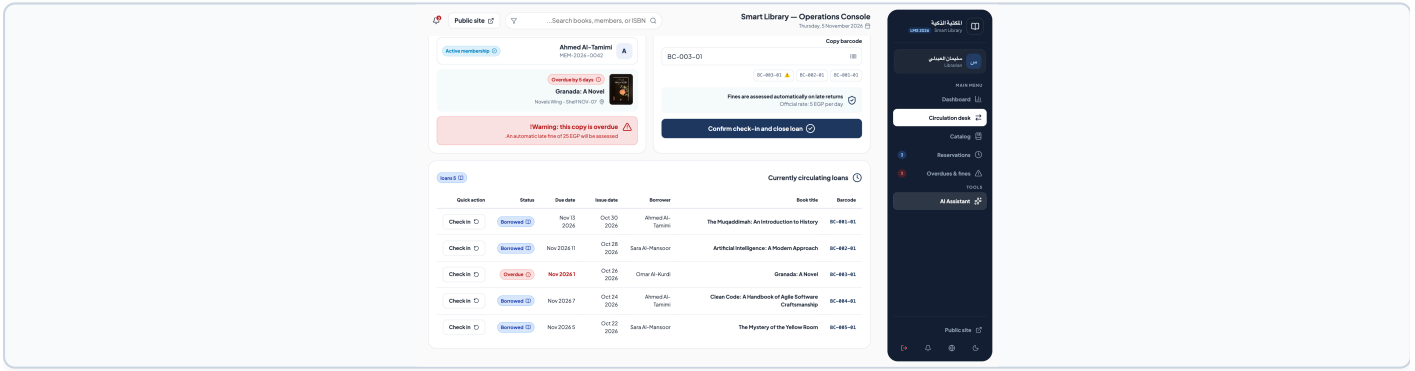
Scanning the item barcode queries the database with `FOR UPDATE` , verifying that the copy is on shelf and not reserved for another patron in the priority queue.

Step 3: Atomic Transaction Completion

Clicking **Complete loan and issue copy** writes the loan record, marks the copy `borrowed` , decrements available count, logs an audit entry, and renders a confirmation toast.

Automated Checkin & Penalty Assessment

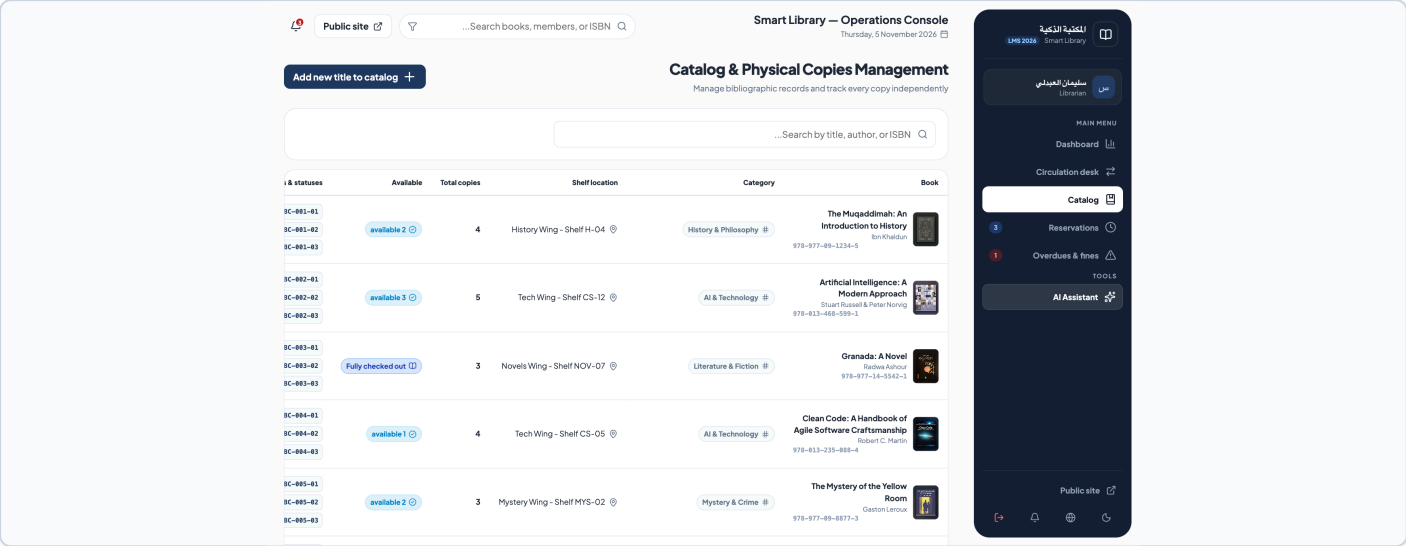
Scanning an overdue copy immediately triggers the late fee calculation engine, alerting staff of the exact fine in Egyptian Pounds (EGP).



UI WALKTHROUGH Master active loans table showing barcode, title, borrower, due date, status badge, and instant check-in buttons.

Walkthrough Stage 8: Bibliographic Catalog Administration

Located at `/librarian/catalog`, this interface allows cataloging librarians to manage master title records, inspect barcode clusters, and provision new acquisitions into physical inventory.



UI WALKTHROUGH Staff catalog management table displaying title metadata, shelf locations, total vs available copies, and barcode clusters.

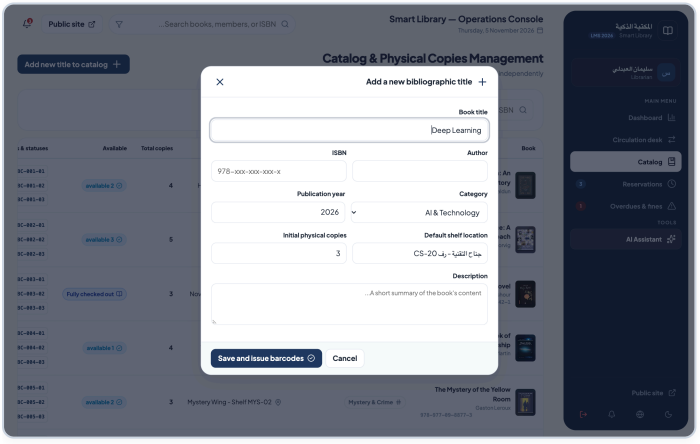
Master Catalog Table

The table organizes all collection records with direct visibility into:

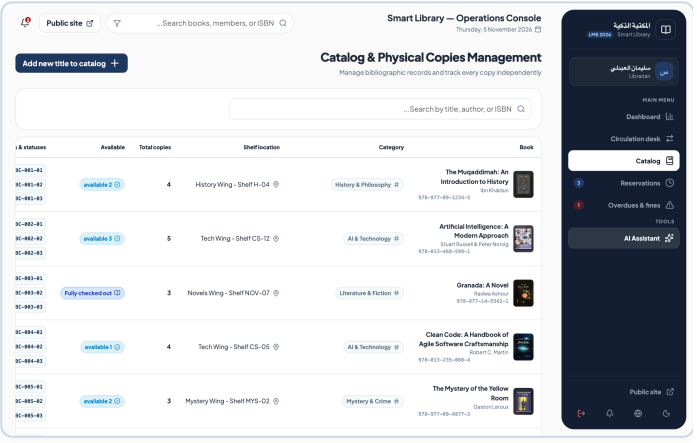
- ✓ Book cover thumbnail, title, author, and ISBN-13.
- ✓ Taxonomy category badge and precise shelf coordinate.
- ✓ Total vs. available copy badges.
- ✓ Barcode cluster badges (e.g. `BC-001-01`, `BC-001-02`).

Automated Copy Provisioning

When adding a new title via the modal, the system accepts an initial copy count (e.g. 3 copies). Upon submission, the database generates discrete inventory rows with sequential barcodes and identical shelf assignments.



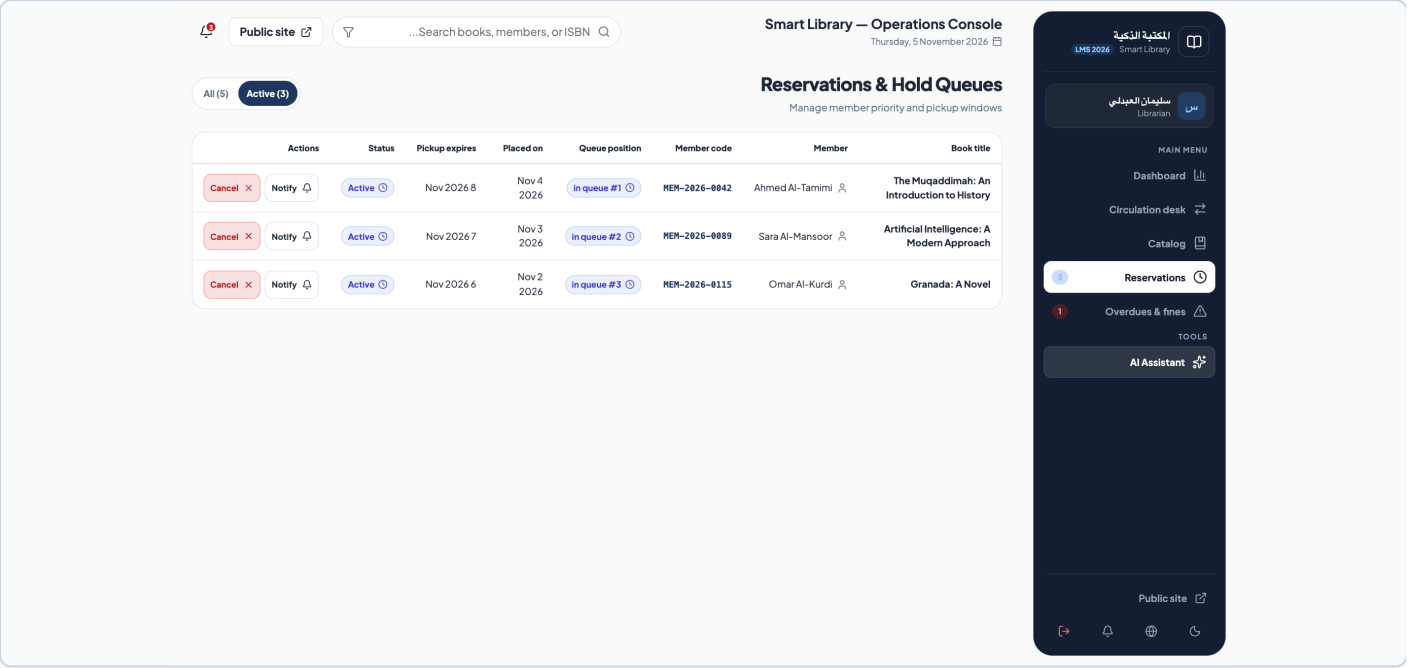
UI WALKTHROUGH Add new bibliographic title modal with fields for ISBN, author, category, publication year, shelf coordinate, and copy count.



UI WALKTHROUGH Staff copy-level status editor allowing inline toggling between available, borrowed, reserved, and maintenance states.

Walkthrough Stage 9: Priority Reservations & Hold Queues

When popular titles are completely checked out, patrons place holds. The `/librarian/reservations` module governs queue order, automated member alerts, and 48-hour pickup window enforcement.



UI WALKTHROUGH Reservations management console displaying title, waiting member, queue position rank, hold placement timestamp, and notification actions.

Hold Queue Mechanics & Workflow

1. Sequential Queue Ordering

Holds are indexed chronologically. Each reservation receives an explicit position rank (`QUEUE #1`, `QUEUE #2`). When a preceding hold is cancelled or fulfilled, subsequent positions automatically shift up.

2. 48-Hour Pickup Expiration Window

When an overdue or loaned copy is checked in, the system alerts the first member in line and activates an expiration countdown (default: 48 hours). If unclaimed, the hold lapses to the next patron.

3. Staff Notification Controls

Circulation staff can trigger notifications manually by clicking **Notify Member**. Once notified, the action button disables to prevent duplicate notifications.

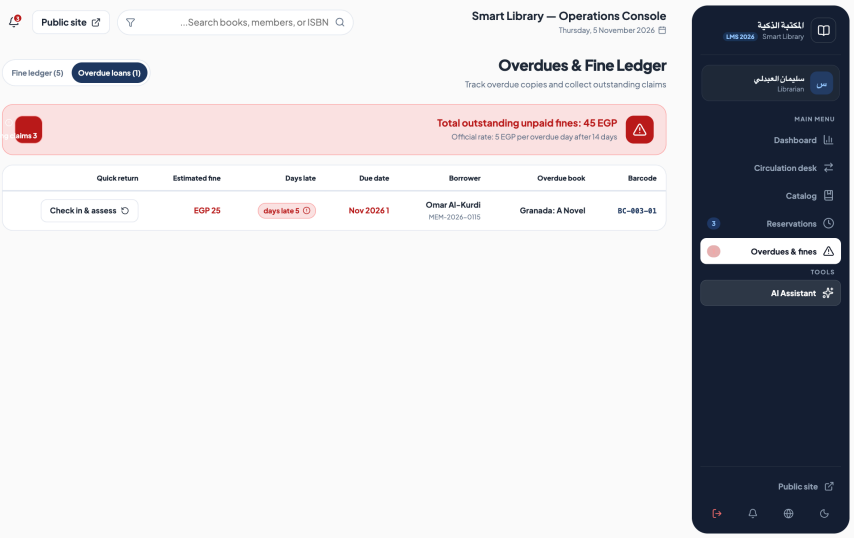
4. Fair Allocation Enforcement

The lending desk restricts checkouts of reserved books to the specific member assigned to Queue #1, protecting patron priority.

> **Equitable Access**
> Dynamic waitlist tracking guarantees transparent, fair access to high-demand academic textbooks without favoritism or manual record-keeping errors.

Walkthrough Stage 10: Financial Overdues & Fines Ledger

Financial transparency is enforced at `/librarian/overdues`. All library fines are officially calculated, tracked, and settled in **Egyptian Pounds (EGP / ج.م)** at the rate of `5.00 EGP per day`.



UI WALKTHROUGH Overdue items tab showing days late counters, calculated penalties in EGP, and one-click 'Check in & assess' action.

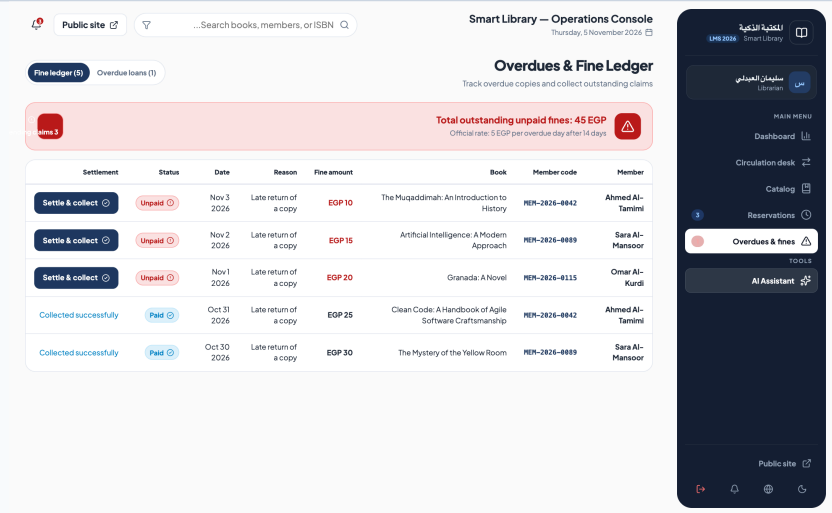
Dual-Tab Financial Operations

Tab 1: Delinquent Active Overdues

Lists every loan exceeding its scheduled return date. Displays the exact days late (e.g. `5 DAYS LATE`) and live penalty amount. Clicking **Check in & assess** returns the copy and formalizes the fine in the financial ledger.

Tab 2: Unpaid & Settled Fines Ledger

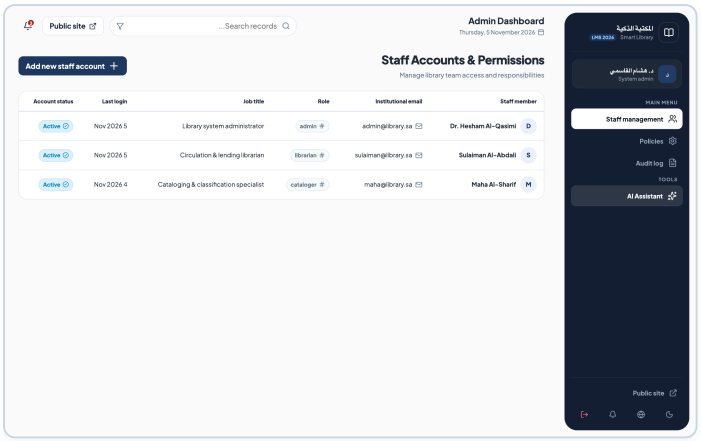
Maintains an immutable historical record of all financial claims. Staff can collect outstanding fees by clicking **Settle & collect**, timestamping the payment and restoring the member's borrowing privileges.



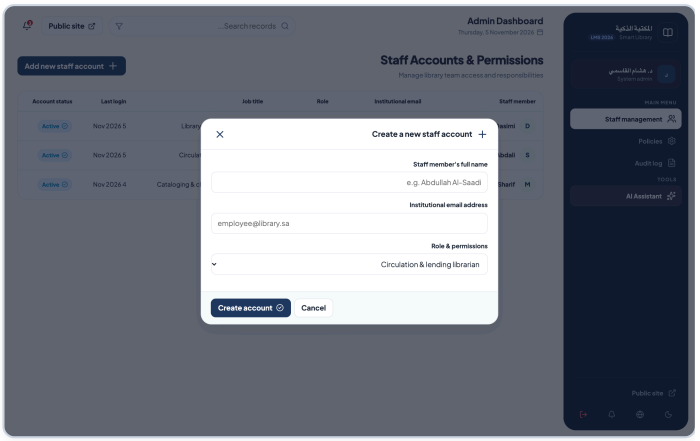
UI WALKTHROUGH Fines ledger tab showing member names, assessed fine amounts in EGP, late return reasons, and settlement action buttons.

Walkthrough Stage 11: Administrative Governance & Policies

The `/admin` suite provides institutional administrators with governance controls over staff accounts, operational circulation parameters, and dynamic policies.



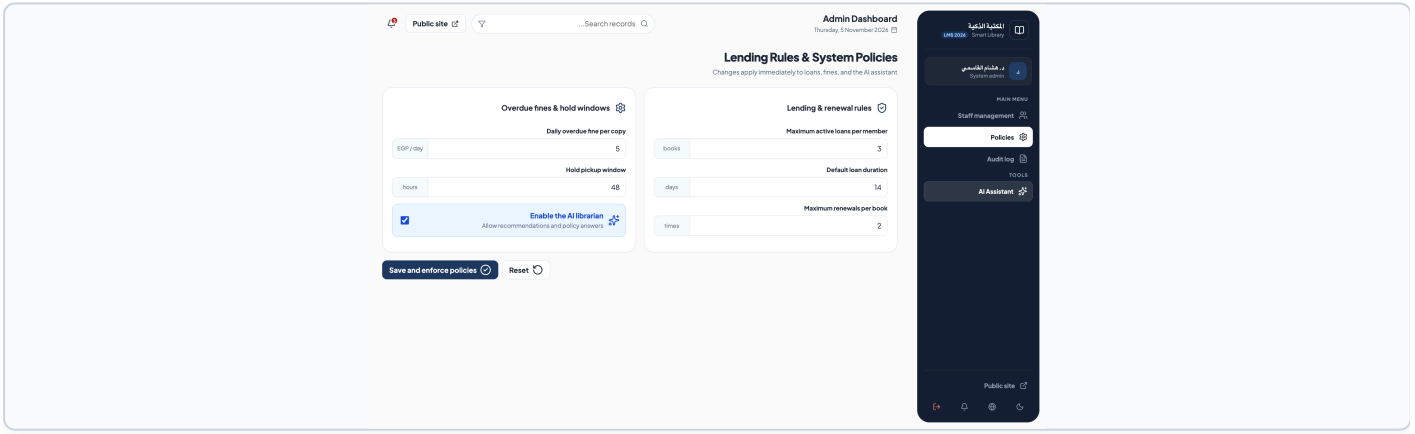
UI WALKTHROUGH Staff administration console listing institutional team members, permissions roles, job titles, and active statuses.



UI WALKTHROUGH Create new staff account modal with fields for employee name, institutional email address, and role selector.

System Operational Configuration Parameters

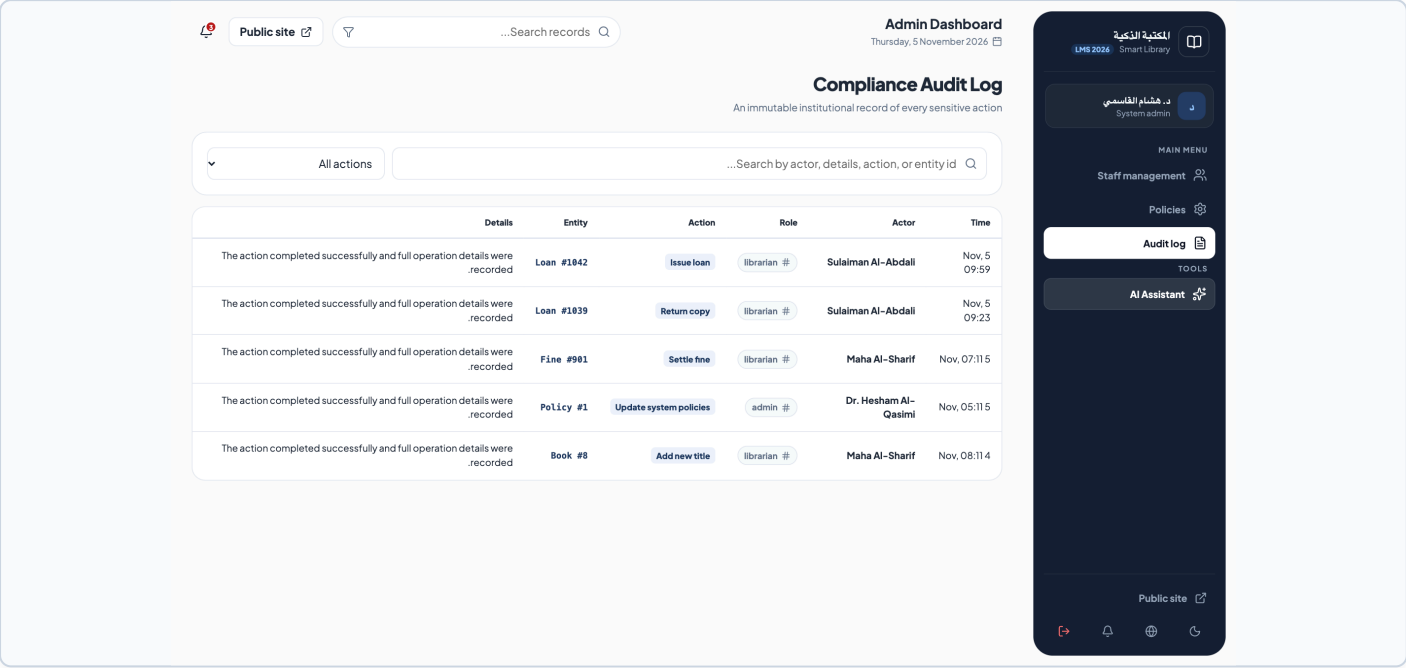
POLICY IDENTIFIER	DEFAULT VALUE	UNIT	SYSTEM ENFORCEMENT SCOPE
`max_borrow_limit`	`3`	Books	Maximum concurrent active loans permitted per member account.
`default_loan_days`	`14`	Days	Standard checkout duration before an item is classified as overdue.
`max_renewals`	`2`	Times	Maximum allowable self-service or staff renewals per loan transaction.
`fine_per_day`	`5.00`	EGP / Day	Financial overdue penalty assessed daily after grace period.
`reservation_expiry_hours`	`48`	Hours	Time window allocated for a member to pick up a reserved copy.
`enable_ai_librarian`	`true`	Boolean	Global toggle enabling or disabling the Google Gemini assistant.



UI WALKTHROUGH System policies configuration panel adjusting max borrowing limits, loan durations, daily overdue fines in EGP, and master AI toggle.

Walkthrough Stage 12: Compliance Audit Log & Security Journal

Smart Library LMS maintains an immutable, chronological journal at `/admin/audit`. Every critical operation is permanently recorded with actor identity, role, target entity, timestamp, and detailed descriptions.



UI WALKTHROUGH Compliance audit log table showing timestamped entries for issue, return, fine settlement, and policy update operations.

Security & Regulatory Compliance Architecture

Immutable Event Capture

The audit log captures events across the entire lifecycle:

- ✓ `audit.actIssue`: Physical loan checkout with barcode and borrower.
- ✓ `audit.actReturn`: Check-in processing and copy status release.
- ✓ `audit.actSettle`: Financial penalty collection in EGP.
- ✓ `audit.actPolicies`: Administrative system parameter modifications.
- ✓ `audit.actAddBook`: New catalog acquisition provisioning.

Live Search & Action Filtering

Administrators can search by actor name (Arabic or English), action type, or entity identifier (e.g. `Loan BC-001-01`) with real-time table filtering.

Tamper-Proof Audit Trail

Logs are append-only; database integrity rules prevent deletion or retroactive modification of historical compliance records.

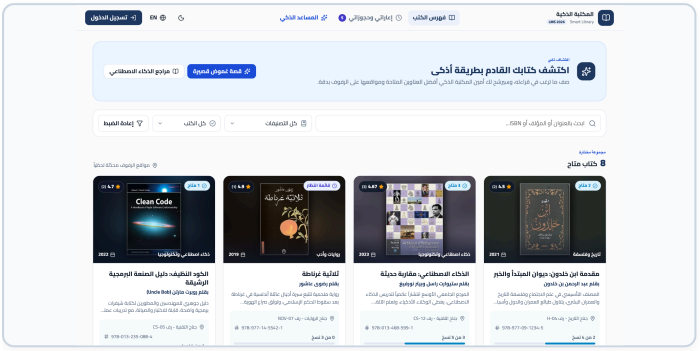
> **Institutional Accountability**
>
> Guarantees full forensic traceability for academic audits, inventory variance investigations, and financial reconciliations.

Walkthrough Stage 13: Internationalization, Dark Theme & Mobile

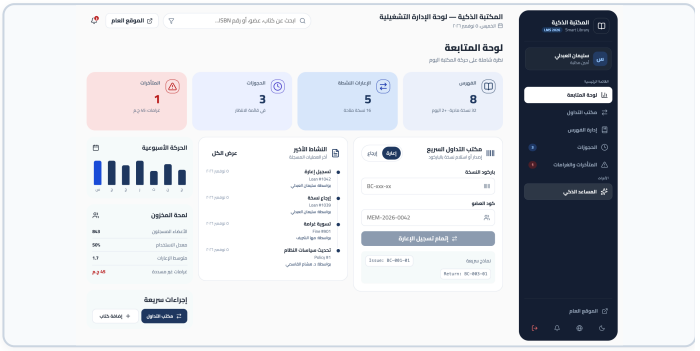
Smart Library delivers flawless ergonomics across languages, color schemes, and device form factors.

1. Native Bilingual Mirroring: Arabic (RTL) & English (LTR)

Instant dynamic direction flipping swaps sidebars, navigation, and typography between *Cairo* and *Plus Jakarta Sans*.



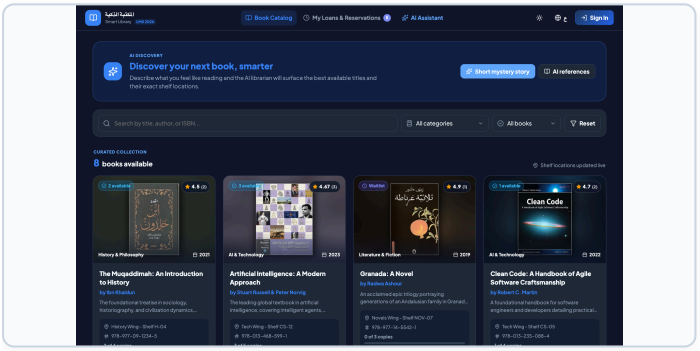
UI WALKTHROUGH Public catalog displayed in native Arabic (RTL) with localized search filters, typography, and card alignments.



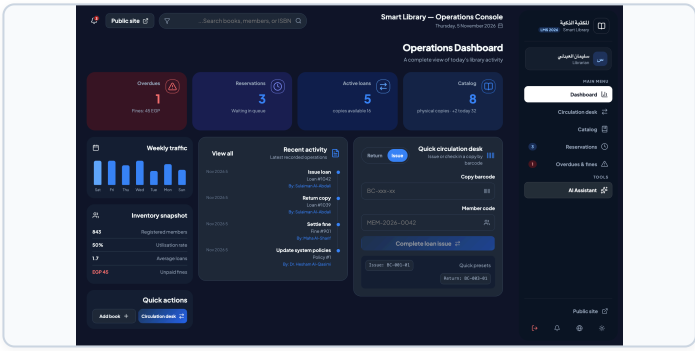
UI WALKTHROUGH Staff dashboard rendered in Arabic RTL with right-hand fixed sidebar and Arabic KPI labels.

2. Curated Midnight Slate Dark Theme

High-contrast dark theme (`#0F172A` page base, `#1E293B` card surfaces) calibrated for low-light library reading rooms.



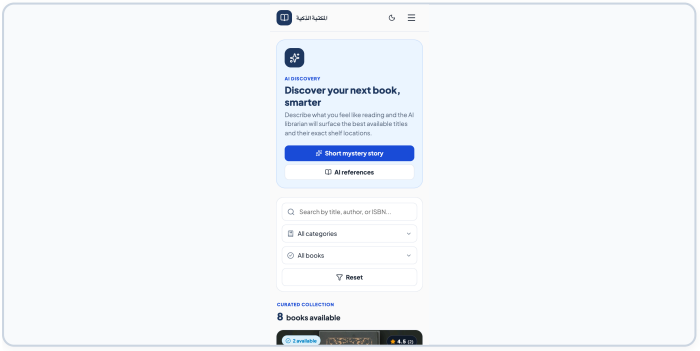
UI WALKTHROUGH Public catalog rendered in Midnight Slate dark mode with glowing Electric Blue accent tokens.



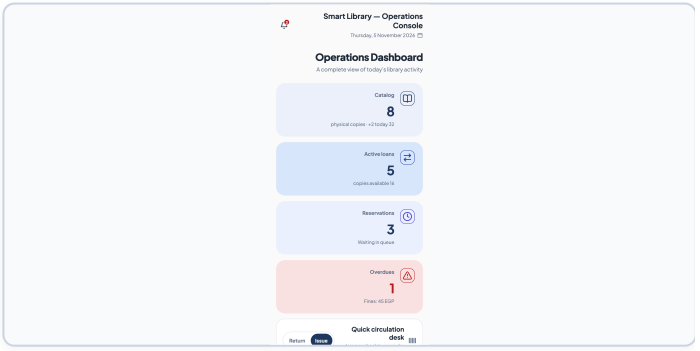
UI WALKTHROUGH Librarian dashboard in dark mode maintaining high contrast ratios and clear data visualization.

3. Responsive Mobile Viewports (iOS & Android)

On mobile (390px), the sidebar transforms into a slide-over drawer, and book grids condense into a touch-friendly feed.



UI WALKTHROUGH Public catalog optimized for mobile screen with collapsible filter drawer and stacked book cards.



UI WALKTHROUGH Staff dashboard on mobile with stacked KPI cards, quick desk touch inputs, and floating action drawer.

Appendix: REST API Endpoints & Database Architecture

The backend architecture exposes a RESTful JSON API partitioned into domain modules, backed by 13 normalized MySQL tables.

REST API Endpoint Catalog

METHOD & PATH	AUTH & ROLE	DESCRIPTION & BUSINESS LOGIC
`POST /api/v1/auth/login`	Public	Authenticates credentials, returns stateless Bearer JWT and user profile.
`POST /api/v1/auth/register`	Public	Creates new patron account, generates member code (`MEM-2026-xxxx`).
`GET /api/v1/books`	Public	Searches catalog with query, category, and availability filters.
`GET /api/v1/books/:id`	Public	Retrieves master title metadata, physical copies list, and reader reviews.
`POST /api/v1/books/:id/ratings`	Member	Submits 1–5 star score and critique, recalculating average book rating.
`POST /api/v1/books`	Librarian+	Creates title record and automatically provisions physical barcode copies.
`POST /api/v1/circulation/loans`	Librarian+	Executes ACID checkout transaction after pre-flight member/copy validation.
`POST /api/v1/circulation/loans/:id/return`	Librarian+	Checks in item, computes late days, assesses penalty in EGP if overdue.
`POST /api/v1/circulation/loans/:id/renew`	Member+	Extends due date by standard duration after verifying renewal policies.
`POST /api/v1/circulation/reservations`	Member+	Places hold on unavailable title and assigns priority queue rank.
`POST /api/v1/circulation/fines/:id/settle`	Librarian+	Settles fine in EGP, timestamps payment, restores borrowing eligibility.
`POST /api/v1/ai/assistant`	Public	Evaluates prompt via Gemini 1.5, matches catalog titles, returns advice.
`GET /api/v1/admin/policies`	Admin	Retrieves current live operational policies and lending parameters.
`GET /api/v1/admin/audit-logs`	Admin	Returns immutable journal of system security, circulation, and admin events.

MySQL Database Schema (13 Normalized Tables)

- ✓ **`users`** : Credentials, email, password hash, role.
- ✓ **`members`** : Member code, active status, max loans.
- ✓ **`books`** : Master title, ISBN, publication year, rating.
- ✓ **`authors`** & **`book_authors`** : Attribution.
- ✓ **`categories`** & **`book_categories`** : Taxonomy.
- ✓ **`book_copies`** : Barcode, shelf location, status.
- ✓ **`loans`** : Issue date, due date, return date, renewals.
- ✓ **`reservations`** : Priority queue rank, expiration date.
- ✓ **`fines`** : Overdue amount in EGP, settlement status.
- ✓ **`book_ratings`** : Star score (1–5) and critique.
- ✓ **`policies`** : Key-value operational configuration.
- ✓ **`audit_logs`** : Immutable actor, role, action log.
- ✓ **`ai_usage_events`** : Privacy-preserving AI metrics.

> **Quick Launch Command:**
> Run `npm run dev` in project root. The Vite auto-backend plugin launches the Express API and MySQL pool automatically.